SIMATS ENGINEERING

CSA17 – ARTIFICIAL INTELLIGENCE LAB

Day 1 – Question 4: 4. Crypt-Arithmetic Problem

Aim: To solve SEND + MORE = MONEY by assigning a distinct digit to each letter.

Python Program

```
from itertools import permutations

letters = "SENDMORY"

for p in permutations(range(10), len(letters)):
    d = dict(zip(letters, p))
    if d['S'] == 0 or d['M'] == 0:
        continue

    send = 1000*d['S'] + 100*d['E'] + 10*d['N'] + d['D']
    more = 1000*d['M'] + 100*d['O'] + 10*d['R'] + d['E']
    money = (10000*d['M'] + 1000*d['O'] +
             100*d['N'] + 10*d['E'] + d['Y'])

    if send + more == money:
        print("SEND =", send)
        print("MORE =", more)
        print("MONEY =", money)
        break
```

Sample Output

```
SEND = 9567
MORE = 1085
MONEY = 10652
```

Result: The program demonstrates the stated Artificial Intelligence problem.